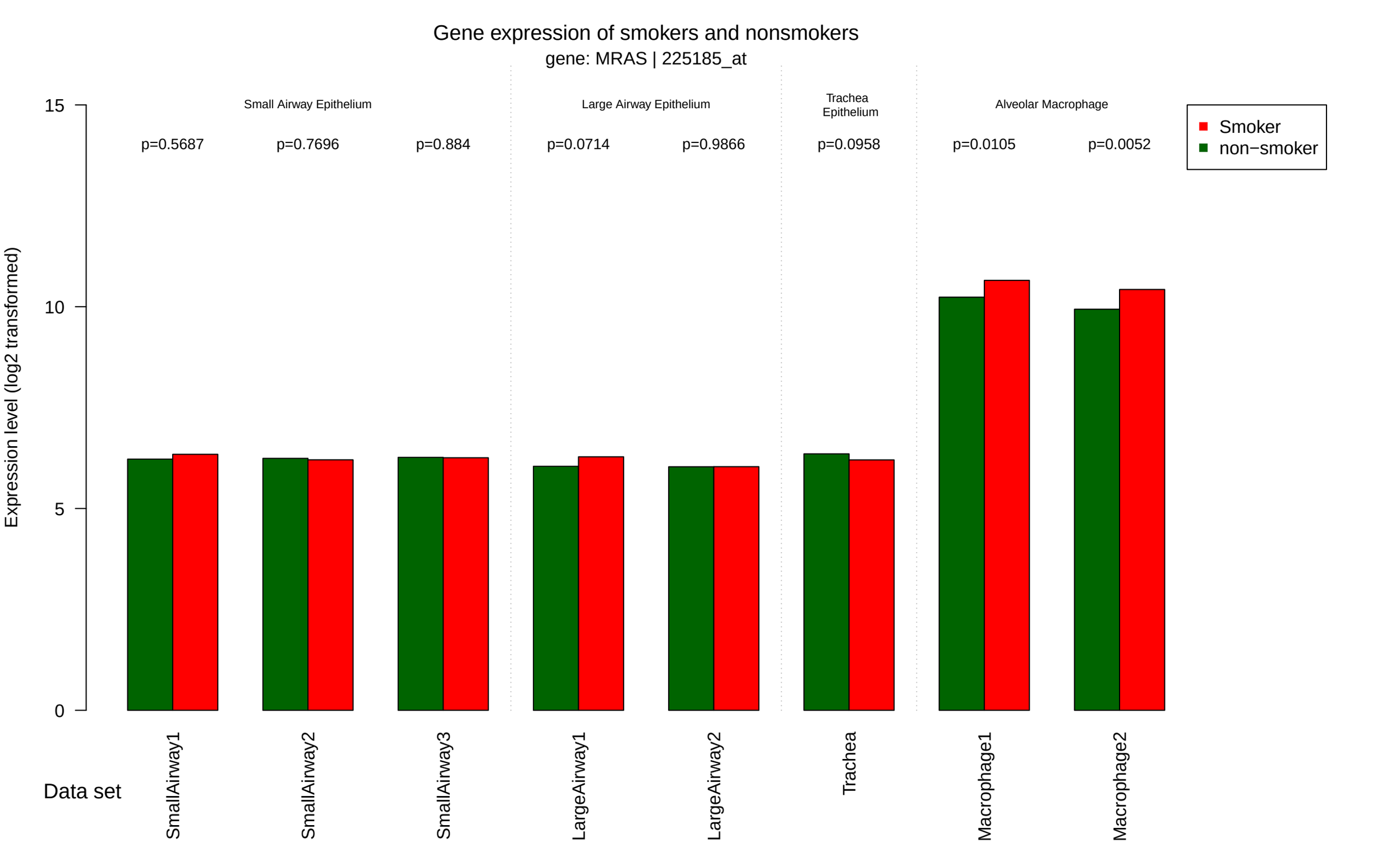
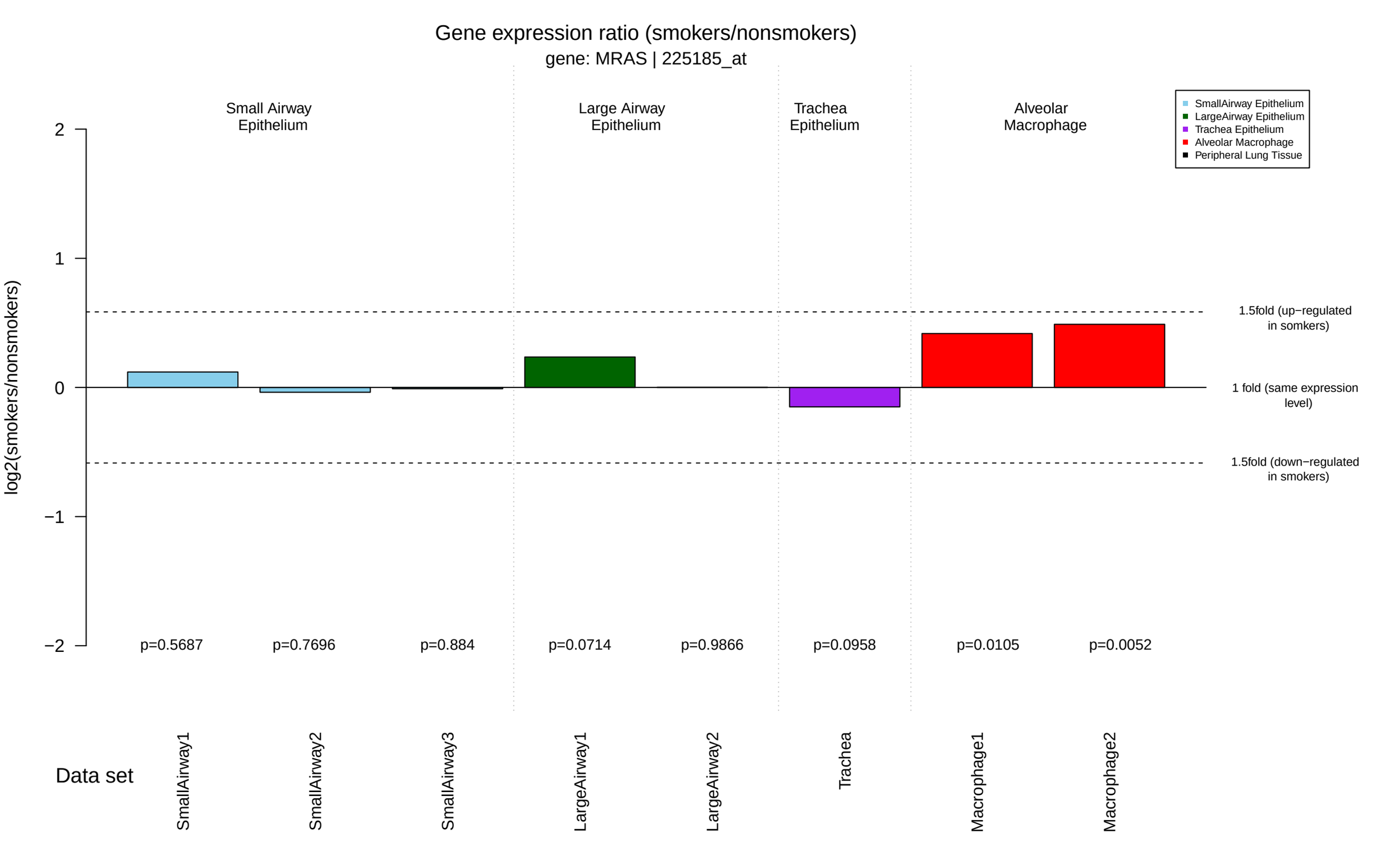
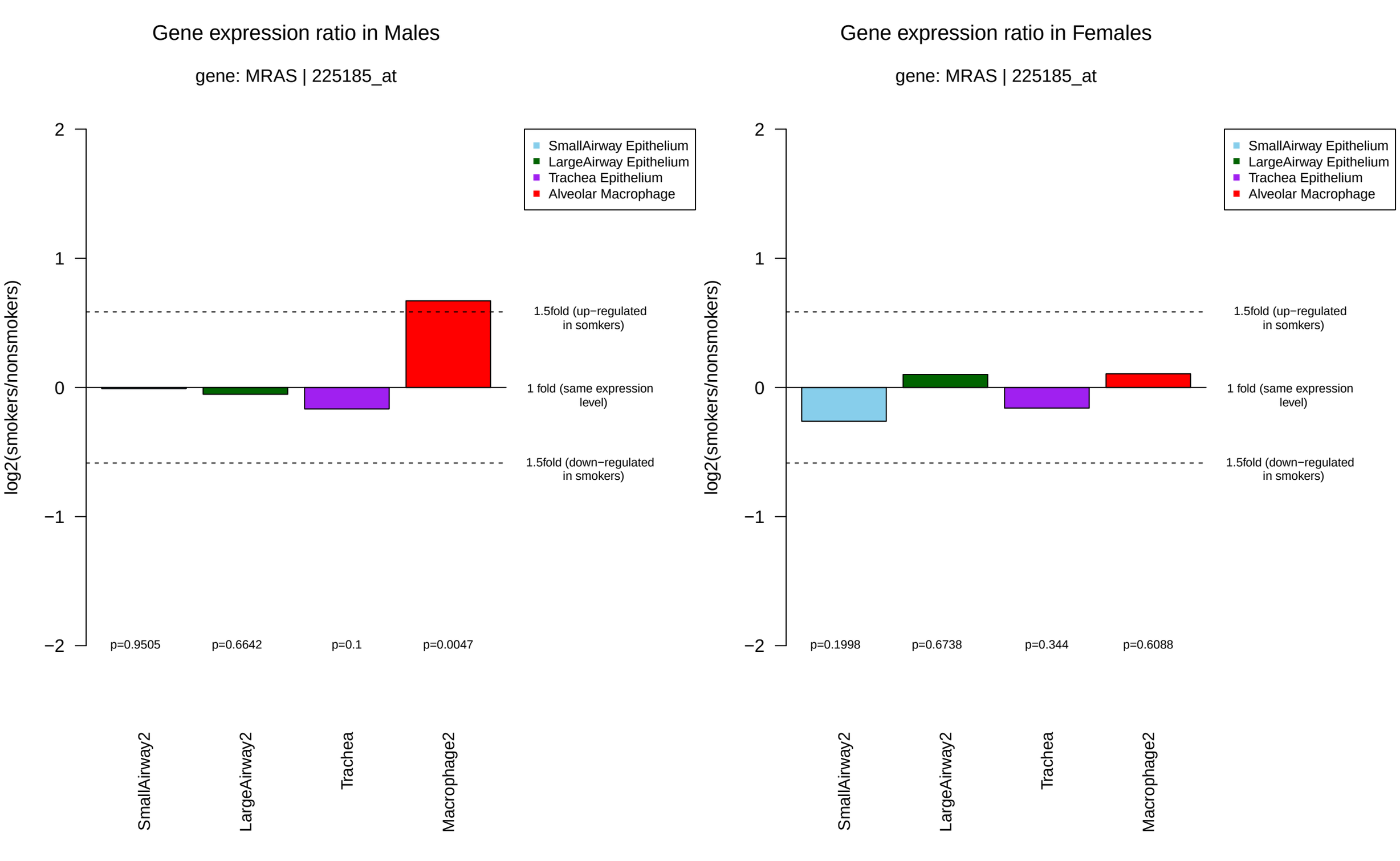
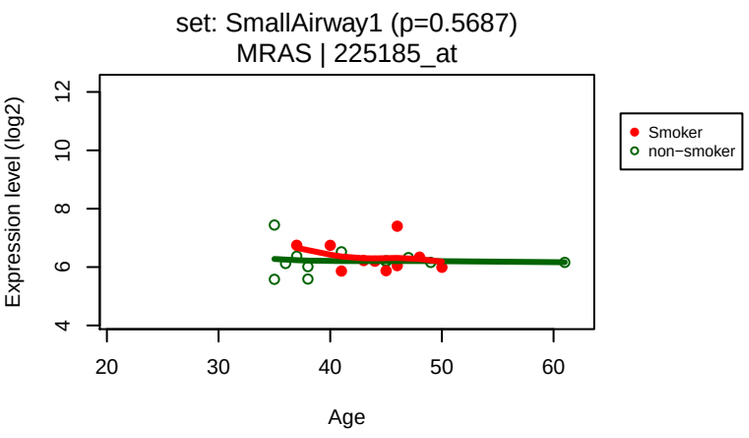
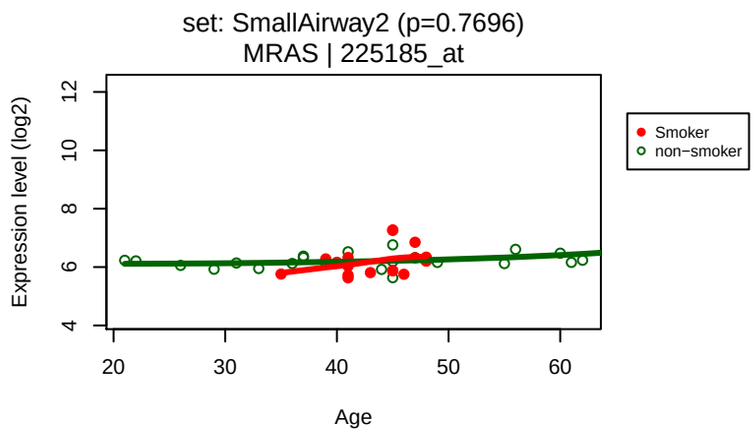
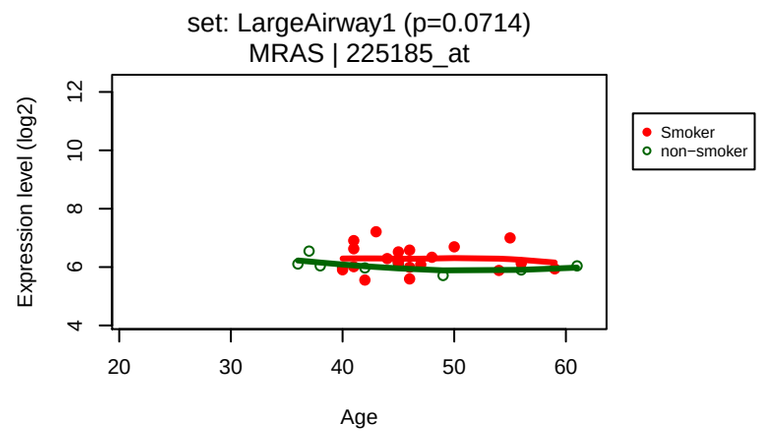

Gene expression of smokers and nonsmokers
gene: MRAS | 225185_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5687
p=0.7696
p=0.884
p=0.0714
p=0.9866
p=0.0958
p=0.0105
p=0.0052
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

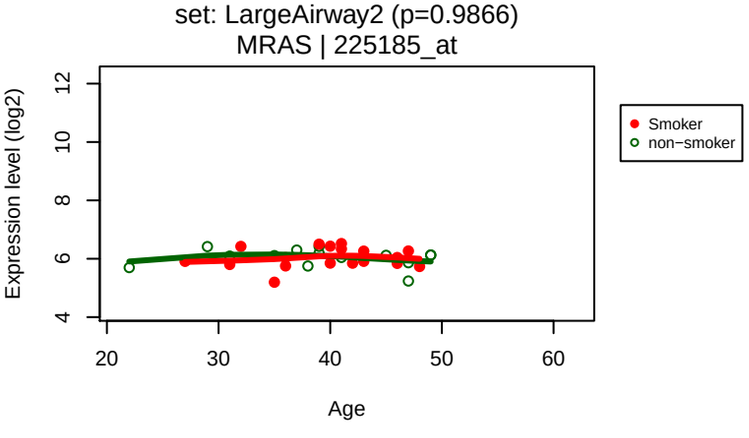

Gene expression ratio (smokers/nonsmokers)
gene: MRAS | 225185_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5687
p=0.7696
p=0.884
p=0.0714
p=0.9866
p=0.0958
p=0.0105
p=0.0052
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

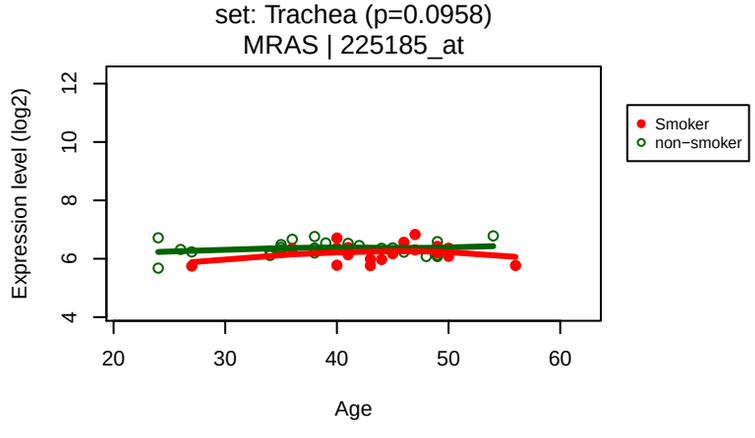

Gene expression ratio in Males
gene: MRAS | 225185_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9505
p=0.6642
p=0.1
p=0.0047
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MRAS | 225185_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1998
p=0.6738
p=0.344
p=0.6088
SmallAirway2
LargeAirway2
Trachea
Macrophage2

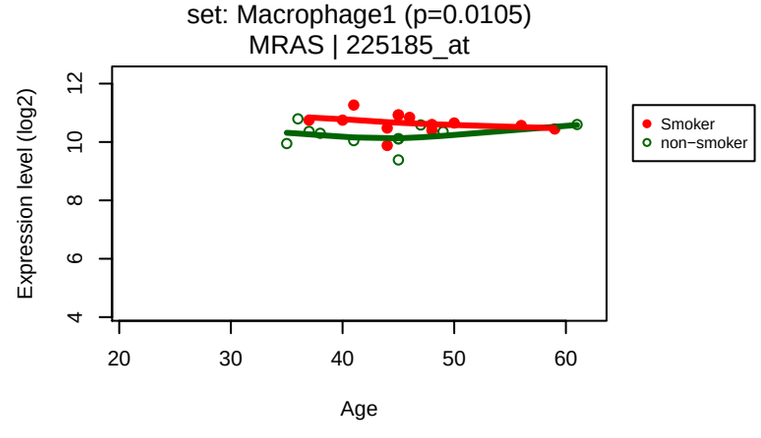

set: SmallAirway1 (p=0.5687)
MRAS | 225185_at
Expression level (log2)
Age
Smoker
non−smoker

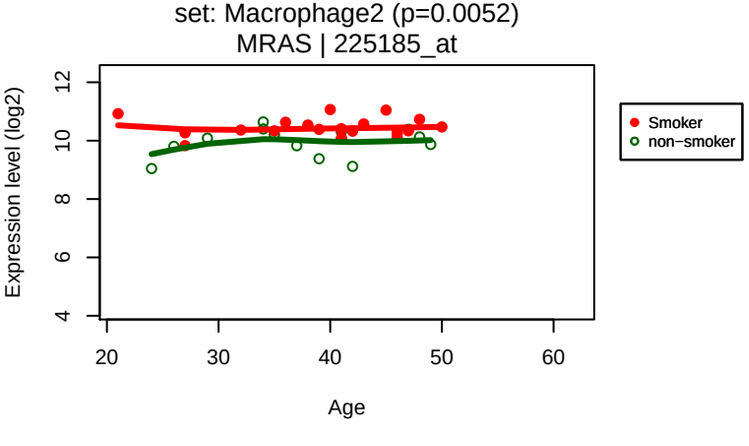

set: SmallAirway2 (p=0.7696)
MRAS | 225185_at
Expression level (log2)
Age
Smoker
non−smoker

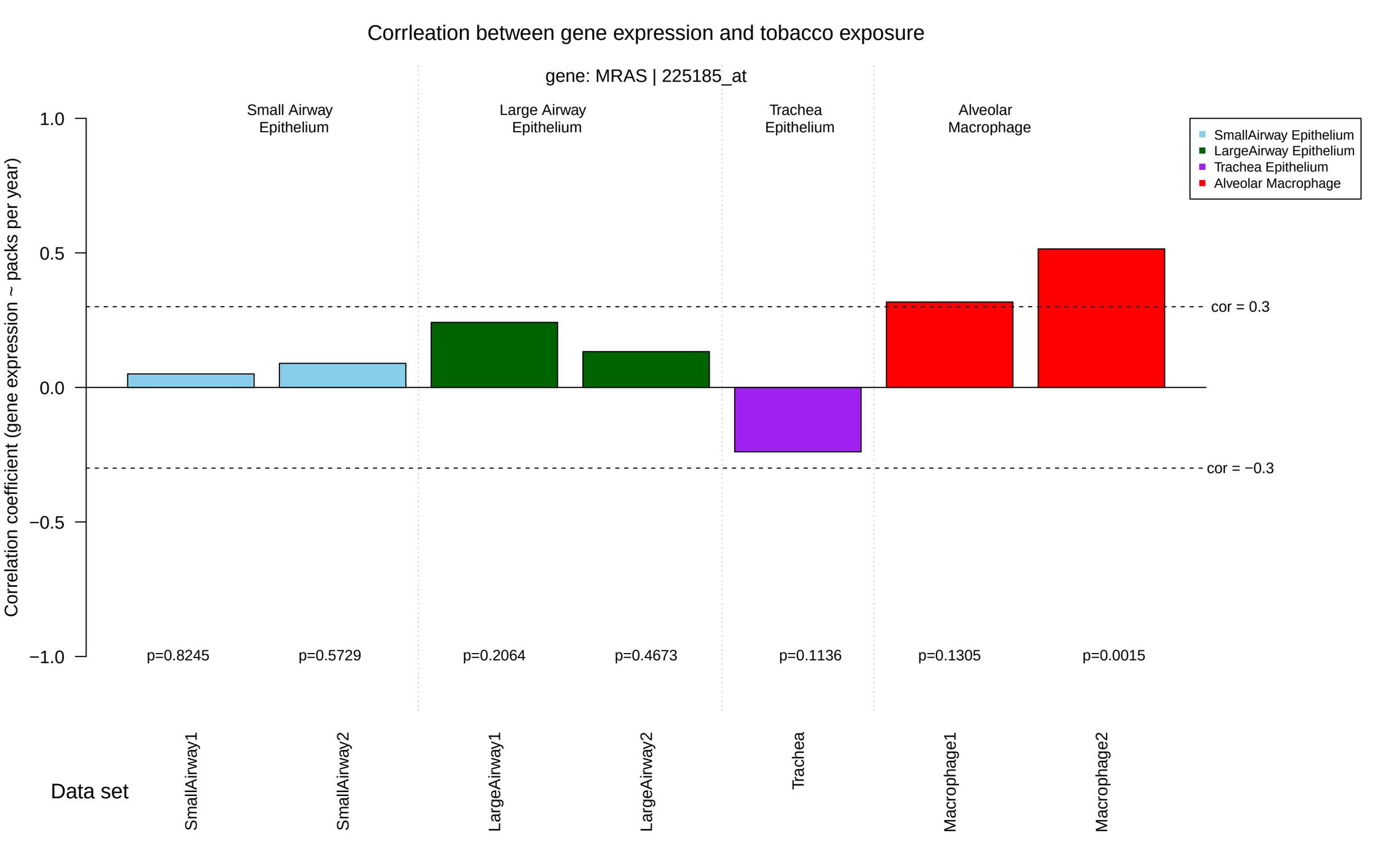

set: LargeAirway1 (p=0.0714)
MRAS | 225185_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9866)
MRAS | 225185_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0958)
MRAS | 225185_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0105)
MRAS | 225185_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0052)
MRAS | 225185_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MRAS | 225185_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8245
p=0.5729
p=0.2064
p=0.4673
p=0.1136
p=0.1305
p=0.0015
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2